

Reciprocity Law

Diaa Al Saleh

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One of the central questions in algebraic number theory is understanding how prime numbers factor in extensions of number fields. In Galois extensions, the splitting behavior of a prime is encoded by its Frobenius element, and in the special case of abelian extensions of $\mathbb{Q}$, this information can be described using one-dimensional representations of the Galois group.

The goal of this paper is to explain how the splitting of prime ideals in finite abelian Galois extensions of $\mathbb{Q}$ can be understood through Dirichlet characters. These characters arise naturally as one-dimensional representations of certain finite abelian groups and provide an explicit bridge between arithmetic properties of primes and the structure of abelian extensions. We begin by introducing Dirichlet characters.

Definition 1. A *Dirichlet character modulo q* is a character of the abelian group $(\mathbb{Z}/q\mathbb{Z})^\times$, i.e a multiplicative homomorphism

$$\chi : (\mathbb{Z}/q\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$$

We say that q is the *modulus* of χ .

Definition 2. The minimum modulus of the Dirichlet character is called the conductor and we denote it by f_χ . A Dirichlet character with modulus f_χ is called *primitive*.

Example 1.

1. Define $\chi : (\mathbb{Z}/5\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$ by $\chi(1) = 1, \chi(2) = i, \chi(3) = -i, \chi(4) = -1$.
2. If p is an odd prime then the quadratic character $(\mathbb{Z}/p\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$ is the Legendre symbol mod p .
3. Let $(\mathbb{Z}/12\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$ be given by $\chi(1) = 1, \chi(5) = -1, \chi(7) = 1, \chi(11) = -1$. Observe that $\chi(a + 3k) = \chi(a)$ so χ is induced by the character $\psi : (\mathbb{Z}/3\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$, where $\psi(1) = 1, \psi(2) = -1$. ψ is primitive so $f_\chi = 3$.

Now let's take an abelian Galois extension $K/\mathbb{Q}$, we want to study $\rho : \text{Gal}(K/\mathbb{Q}) \rightarrow \mathbb{C}^\times$, that is the 1-dimensional Galois representations. We will see that χ will play a central role in describing these representations. Recall that if $K = \mathbb{Q}(\zeta_n)$ then $\text{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$ so in this case the 1-dim dimensional Galois representations can be realized as Dirichlet characters.

Theorem 1. (Kronecker Weber). Every Abelian extension over $\mathbb{Q}$ lies in a cyclotomic extension.

Now suppose $L/\mathbb{Q}$ is abelian with conductor n then by Kronecker-Weber we get a surjective homomorphism

$$\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \twoheadrightarrow \text{Gal}(L/\mathbb{Q})$$

Now suppose p is a prime with $p \nmid n$ so that $L/\mathbb{Q}$ is unramified above p . From algebraic number theory we know that we have a decomposition group $G_p \subseteq \text{Gal}(L/\mathbb{Q})$ and since there is no ramification G_p is generated by Frob_p . Now we can take the map $p \mapsto \text{Frob}_p$ and extend it to a homomorphism from I_n to $\text{Gal}(L/\mathbb{Q})$ where I_n is the subgroup of $\mathbb{Q}^*$ generated by all primes not dividing n . The map $\theta : I_n \rightarrow \text{Gal}(L/\mathbb{Q})$ is called the *Artin* map.

Really the main potato of this is that θ factors through the map $(\mathbb{Z}/n\mathbb{Z})^\times \rightarrow \text{Gal}(L/\mathbb{Q})$ so in the end we obtain the following diagram:

$$\begin{array}{ccc} I_n & \xrightarrow{\theta} & \text{Gal}(L/\mathbb{Q}) \\ \downarrow & & \downarrow \rho \\ (\mathbb{Z}/n\mathbb{Z})^\times & \xrightarrow{\chi} & \mathbb{C}^\times \end{array}$$

Finally we get the reciprocity law $\boxed{\chi(p) = \rho(\text{Frob}_p)}$.

Now we look at some applications of this result. As a habit of me I like to start any examples with my favorite ring i.e $\mathbb{Z}[i]$.

Example 2. Let $\chi : (\mathbb{Z}/4\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$ given by $\chi(1) = 1, \chi(3) = -1$. Now the reciprocity law tells us that when $p \equiv 1 \pmod{4}$ then $\chi(p) = \rho(\text{Frob}_p) = 1$ so it is split in this case. Similarly for $p \equiv 3 \pmod{4}$ we get that it is inert.